

Crop plan audit

Verdict: the scheduling engine passes its arithmetic checks, but this PDF needs revision before it serves as the team's standalone field plan. It mixes first-year and established-year expectations, omits designed food sources, and loses information needed to carry out the jobs.

Reviewed: all 30 pages of the attached export, current main code, 270 regional scenario combinations, 16 generated regional PDFs, targeted tree/animal probes, and the live sample planner. Audit only: no source changes or deployment.

Priority	Finding	Consequence
High	Avocado uses an 11-month national union	The display is not Ubhejane's local picking season.
High	Banana disappears when months are unsourced	A crop already drawn on the design is absent without explanation on paper.
High	Honey is lost from the print pipeline; coop needs a hidden choice	The export can omit both hives and poultry. Simple mode hides the poultry setup.
High	Cover and food chart mix time modes	12 fresh months and 11 stored months do not describe the reconstructed first year.
High	Monthly field sheets omit crop area shares	A worker cannot tell which third/half of a bed to plant from the job sheet.
Medium	Oats has no buying row; storage conditions are omitted	Required inputs and the conditions behind stored-food claims are missing.
Medium	Missed starters become existing automatically	A skipped September sowing can still generate a November harvest.
Low	Tiny type, orphan continuation pages, incomplete record fields	The document is harder to print, read and use in the field.

What did pass

The printed 508.3 m2 area, 618.7 kg crop-cycle benchmark, 1.22 kg/m2 density, 464.0 kg after the confirmed 25% loss allowance, and all 12 workload counts reproduce at displayed precision. No bed overbooking was found in the reconstructed 71 rows or the 270 generated scenarios. These are planning calculations; they do not verify actual harvests.

Current main checked: f76814b2aa25. Node 24.19.0: typecheck clean; 4,379 passing tests, zero failures, one existing TODO. git diff --check clean. The TODO concerns concurrent design-shape loss and is disclosed on page 9.

Avocado and banana

Avocado: your long-season observation is correct

One placed avocado is shown in February through December, excluding only January. `sourcedSeasonMonths` unions every sourced window; `buildTreeAvailability` has no site or cultivar input. The same charted season therefore follows the tree into every location tested.

Record used by the code	Months
General South African industry	February-October
Hass, warm subtropical regions	June-October
Hass, cool subtropical regions	August-December
Union displayed for each placed avocado	February-December (11 months)

Those windows are sourced, but combining them does not establish an individual tree's season. The KZN DARD guide distinguishes Fuerte (May-November), Hass/Edranol (August-December), Pinkerton (July), and Ryan (November-February). These are reference examples, not a replacement season selected for Ubhejane. The actual cultivar, local conditions and observed bearing must decide that. [1, 2]

Banana: recognised, then filtered away

Placed item	Resolved species/group	Calendar result
banana_clump	Existing banana record	No months; filtered out
banana_circle without speciesId	No unambiguous species mapping	No group; filtered out
banana_circle with explicit banana speciesId	Explicit species wins	Still no sourced months

The banana dossier's harvest months were deliberately emptied in its 28 September evidence review. That avoids inventing a year-round calendar, but `lib/perennial-harvest.ts:207` then hides the plant entirely. The PDF also lacks the on-screen list of trees with unknown picking months. The existing regional PDF harness contains no banana, so it did not exercise this case.

Required change: retain the placed banana in an inventory or an undated row, with the reason no bar is drawn. Keep unknown timing explicit. For avocado, select a justified regional/cultivar window or show an undated local season until confirmed.

KZN DARD gives banana plant-to-harvest intervals of 13-20 months depending mainly on temperature and cultivar, and describes strong water and cold limitations. A newly proposed banana cannot be treated as food already available. No fixed Ubhejane monthly banana schedule was invented in this audit. [1]

Hives and the coop

Both built-in elements are mapped: **beehive** -> **bee** and **chicken_coop** -> **chicken**. Placing them identifies housing, while a separate per-design choice specifies the enterprise. The mapped structure count is not an animal count. [lib/animal-enterprises.ts:136](#) and [app/facilitator/crops/page.tsx:1120](#).

Focused fixture: 2 existing hives + 1 existing coop	On-screen/data result	PDF result
No enterprise choices	No dated food; setup note	No animal row or setup explanation
Honey chosen	Undated honey line and local-flow explanation	Honey absent
Honey + commercial layers chosen	Undated honey + 12 egg slots	Eggs present; honey absent
Village chickens or broilers chosen	No verified monthly windows	Their produce absent
Proposed commercial layers, from-today mode	No current produce	Correctly excluded

Two confirmed implementation problems

- 1. Simple mode hides setup.** The calendar says to choose an enterprise under Animals on your map, but that card sits inside the !simple block. The control cannot be reached from that view. [app/facilitator/crops/page.tsx:2131](#) and [app/facilitator/crops/page.tsx:2190](#).
- 2. Honey is lost between screen and paper.** `unmarkedAnimalLines` deliberately keeps honey as a line without a month bar. `printableAvailability` receives only the dated animal slots, so the PDF cannot carry that line. A focused PDF export with both Honey and Laying hens selected printed only the hens.

The PDF uses the current `includeAnimals` switch. Commercial layers have a sourced all-year window tied to managed lighting; indigenous chickens and broilers do not have verified monthly windows. Selecting commercial layers merely to force an icon would misdescribe a village coop. The printed display must preserve those conditions and unknowns. [3]

What remains specific to your saved design

I have not read Ubhejane's private saved enterprise choices, `includeAnimals` switch, item status, or selected canvas state. The code paths above reproduce the omission, but do not prove which saved setting explains your coop. The exact design should show its detected hives and coop, the purpose chosen for each, and a reason beside any product without a dated bar.

Acceptance: a placed hive remains visible with local-flow timing unknown; every detected coop has an accessible purpose choice in Simple mode; export retains both dated and undated entries. The PDF should report unknown production rather than silently implying absent infrastructure.

The year-one contradiction

Measure in this export	Printed/dashboard	Reconstructed first year
Fresh-picking months	12 / 12	11 / 12; October has none
Stored-food months	11 / 12	6 / 12; April-September
October field space used	78% on food chart	32% in dated first-year schedule
November field space used	78% on food chart	55% in dated first-year schedule
December field space used	99% on food chart	76% in dated first-year schedule

Page 6 does say AN ESTABLISHED YEAR, which makes that chart an intentional scenario. Its large title nevertheless dates it Oct 2026-Sep 2027, alongside a land calendar and field sheets for the actual first year. Page 1's 12/12 fresh-month claim is also inconsistent with page 17's zero October harvest jobs.

The dashboard calls `buildFoodAvailability` without a rolling horizon. That calendar-mode call can wrap a future cohort's harvest into a month before its first sowing. Passing a dated 12-month horizon produces the 11/12 and 6/12 figures above. The established-year food chart separately removes existing and one-off starter rows, so it represents a different cohort set.

First-year schedule reproduced from the 71 printed rows

Month	Fresh kinds	Stored kinds	Mapped space used
Oct 2026	0	0	32%
Nov 2026	2	0	55%
Dec 2026	2	0	76%
January 2027	1	0	77%
February 2027	3	0	78%
March 2027	8	0	77%
April 2027	5	2	79%
May 2027	7	2	78%
June 2027	6	2	58%
July 2027	5	3	58%
August 2027	5	3	77%
Sept 2027	5	3	78%

Counts above mean distinct crop kinds with a modelled window, not kg, portions, nutritional adequacy or actual crop observations. Existing status for the seven September starters was inferred from the saved note and their absence from the shopping list. Confirm those crops were actually planted before relying on their November/December harvests.

Required change: make the default dated export consistently read from today. Print an established annual template separately and label its repeating-year assumptions at every headline. Keep 618.7 kg labelled as a crop-cycle benchmark: some included cohorts finish after September 2027, including Plot 4 potatoes sown in August.

Locations and climates tested

I reran the existing regional PDF harness for eight sites, irrigated and rain-fed, using its stored NASA POWER climatology fixtures and 2 October 2026 as the start date. I added the recorded Mkuze demo-site climate from the existing site-climate test, then swept all 12 start months. October also exercised whole-bed family, commercial and hybrid options: **270 scenario combinations** total.

Site / fixture	Calendar class	Plantings I / R	Year-one fresh months I / R
Durban coast	mild-frost	58 / 38	10 / 5
Mthatha	summer	60 / 53	10 / 9
Bloemfontein	summer	60 / 32	10 / 6
Johannesburg	summer	60 / 31	10 / 5
Pietermaritzburg	summer	60 / 51	10 / 9
Tzaneen coordinate	summer	60 / 31	10 / 5
Kimberley	summer	57 / 9	10 / 1
Stellenbosch	winter	59 / 34	10 / 4
Mkuze recorded demo climate	mild-frost	64 / 32	10 / 3

I = reliable irrigation selected; R = rain-fed. These are newly generated fixture plans with a common crop selection and an empty starting plan, not replacements for the attached 71-row plan. The eight NASA fixture records cover 2001-2020 normals. The Mkuze record was captured on 19 August 2026; it was not refreshed from live weather during this audit.

Independent checks and checks that can fail

A separate interval calculation reconstructed occupied bed shares for 36 months, including annual recurrence, one-time starters and nursery reservation. All 270 plans stayed at or below one physical bed, respected their configured sowing calendars, stayed inside the code's upper heat limits, and avoided rain-fed months rejected by its water gate. No crop outside the selected list and no partial-area staple plot appeared.

Deliberately invalid area shares and overlapping whole-bed crops withheld the benchmark. A rain-fed request without usable site climate produced no automatic plan and an explanation. The checks therefore exercise failure cases, not just the engine's current outputs.

What this establishes, and its practical limit

The engine is responsive to rainfall and irrigation; the regional audit was worthwhile. Fruit seasons remain identical across sites because their builder has no location input. Crop heat checks use monthly means and the rain check is a coarse rainfall/PET screen. Neither observes local frost pockets, dry spells, soil water storage, irrigation capacity, cultivar choice or pests. Passing these rules cannot certify that a particular crop will succeed on the ground.

Can the team carry it out?

Area shares vanish from the monthly job sheets - high priority

The full plan allocates many crops to one third or half a bed. October's field sheet instead says Sow carrots - Bed 2 and Sow green beans - Bed 3, with spacing but no fraction, square metres, strip ID or matching map. These beds have other cohorts sharing them. [lib/crop-export-benchmark.ts:596](#) groups work by bed and crop, dropping the allocation. A worker following just the monthly sheet can sow the whole bed and create the conflict the arithmetic carefully avoids.

Carry each crop's area/fraction and a stable sub-bed reference through to every sowing and transplant instruction. Keep the exact saved geometry unchanged. Include a matching bed/plot map or a clear field label scheme; no location map is present in this PDF.

Oats is missing from procurement - medium priority

Oats is scheduled in Bed 3 in March and April and across Plot 1 in April, but it has no buying-list entry. [lib/crop-plan.ts:1643](#) intentionally skips all kg/ha seed-rate crops because the buying model handles pieces and packet guidance. Keep a rate-based shopping row, even when the exact purchase quantity still requires confirmation.

The calendar also stars oats in Bed 3 in September, while its legend says a star means picking. The field-sheet logic correctly treats oats as a cover to terminate. The calendar should use a distinct cover-management marker. [lib/crop-export-benchmark.ts:478](#) and [lib/crop-export-pdf.ts:941](#).

The buying list is a sourcing guide, not a completed order

Direct-sown entries say Packet rate needed. Printed numbers are final plant positions, not seed counts; losses and germination have not been converted into orders. For tray crops the row is staged at the earliest transplant month, with a note to source seed before the earlier nursery month. For October cabbage/tomato nursery work, that note sits in November's buying section. If the team raises its own trays, add an explicit seed-source row before the nursery job or a visible linked choice between seed and ready-grown seedlings.

Storage claims lose their conditions - high priority

Six crops have storage durations and detailed conditions in the catalogue, but none of those conditions is printed in the 30-page plan. The PDF says stored under named conditions without naming them. A stored-food window assumes curing, dry grain or suitable temperature/ventilation depending on the crop; those requirements materially affect whether the food exists months later. Export the existing source-backed conditions beside each duration rather than suggesting new values.

Other planning inputs still absent

The 464.0 kg benchmark after losses does not establish enough meals for the creche. It has no validated children/portion demand, monthly kg curve, recorded labour hours or irrigation supply budget. Workload counts jobs, not effort. These are missing decision inputs, not reasons to invent demand or water numbers. Soil-test upload language on page 2 also overstates the crop planner: its suggestion inputs contain climate and choices, not laboratory soil values that rebuild this crop schedule.

Visual and print audit

All 30 original pages were rendered and visually inspected. No clipped text, broken table borders, black glyph boxes or out-of-page words were found. Headers, crop grouping, icons and repeated column headers are consistent. The main visual problems are readability and how information splits across pages.

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LAND OCCUPANCY

LAND OCCUPANCY

Bed calendar: Oct 2026 - Sep 2027

BED / PLOT		Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
Bed 1	9.0 m²	Cu 1/2	Cu 1/2	Cu* 1/2	Ca 1/3 To 1/3	Ca 1/3 TS 1/3	Ca 1/3 To 1/3	Ca 1/3 To 1/3 TS* 1/3	Ca 1/3 To* 1/3	Ca 1/3 Ca* 1/3 To* 1/3	Ca 1/3 To* 1/3	Ca 1/3 Pe 1/3	Ca 1/3 Car 1/3 Pe 1/3
Bed 2	9.0 m²	Car 1/3 GB 1/3	Car 1/3 GB* 1/3	Car 1/3 SP 1/3	Car 1/3 SC 1/3 SP 1/3	Car* 1/3 SC 1/3 SP 1/3	SC 1/3 SC* 1/3 SP 1/3	SC 1/3 SC* 1/3 SP 1/3	SC 1/3 SC* 1/3 SP* 1/3	SC* 1/3 SC* 1/3 SP* 1/3	Ca 1/3 SC* 1/3 SC* 1/3	Ca 1/3 SC* 1/3	Ca 1/3 GB 1/3 SC* 1/3
Bed 3	9.0 m²	GB 1/3 TS 1/3	Car 1/3 GB 1/3 TS* 1/3	Car 1/3 GB* 1/3 GB 1/3	Car 1/3 GB 1/3	Car 1/3 GB* 1/3	Car 1/3 Car* 1/3 Ca 1/3	Car 1/3 Oa 1/3 Oa 1/3	Car 1/3 Oa 1/3 Oa 1/3	Car 1/3 Oa 1/3 Oa 1/3	Car* 1/3 Oa 1/3 Oa 1/3	Ca 1/3 Oa 1/3 Oa 1/3	Ca 1/3 Oa* 1/3 Oa 1/3
Bed 4	9.0 m²	GB 1/2	GB 1/3 GB* 1/2	Co 1/3 GB* 1/3 Pep 1/3	Co 1/3 GB* 1/3 Pep 1/3	Co 1/3 Pep 1/3	Co* 1/3 On 1/3 Pep 1/3	On 1/3 Pep* 1/3	On 1/3 Pep* 1/3	On 1/3 Ka 1/3 Pep* 1/3	Ka 1/3 On 1/3 SC 1/3	Ka 1/3 On 1/3 SC 1/3	Ka 1/3 On 1/3 SC* 1/3
Bed 5	9.0 m²	Cu 1/3 SP 1/3	Cu 1/3 SP 1/3	Cu* 1/3 GB 1/3 SP 1/3	GB 1/3 GB 1/3 SP 1/3	GB* 1/3 GB 1/3 SP 1/3	GB* 1/3 SP* 1/3	Be 1/3 On 1/3 SP* 1/3	Be 1/3 On 1/3	Be 1/3 Co 1/3 On 1/3	Be* 1/3 Co 1/3 On 1/3	Be* 1/3 Co* 1/3 On 1/3	Ca 1/3 On 1/3
Bed 6	9.0 m²	TS 1/2	TS* 1/2	GB 1/3	GB 1/3 GB 1/3	GB 1/3 GB* 1/3	GB* 1/3	On 1/3 TS 1/3	On 1/3 TS 1/3	Co 1/3 On 1/3 TS* 1/3	Co 1/3 Co 1/3 On 1/3	Co 1/3 Co* 1/3 On 1/3	Co* 1/3 On 1/3 To 1/3
Bed 7	9.0 m²		Ca 1/3 SP 1/3	Ca 1/3 SP 1/3	Ca 1/3 SP 1/3	Ca 1/3 SP 1/3	Ca 1/3 SP* 1/3	Ca* 1/3 SP* 1/3	Ca* 1/3 SP* 1/3	Car 1/3 Le 1/3 Pe 1/3	Car 1/3 Le 1/3 Pe 1/3	Car 1/3 Le 1/3 Pe 1/3	Car 1/3 Le 1/3 Pe 1/3
Bed 8	9.0 m²		To 1/3	Ca 1/3 To 1/3	Ca 1/3 To 1/3	Ca 1/3 To 1/3	Ca 1/3 To* 1/3	Ca 1/3 To* 1/3 TS 1/3	Ca 1/3 To* 1/3 TS 1/3	Ca* 1/3 To* 1/3 TS* 1/3	Car 1/3 Le 1/3 Pe 1/3	Car 1/3 Pe 1/3	Car 1/3 Pe 1/3
Bed 9	9.0 m²	GB Full	GB* Full	Cu 1/3 To 1/3	Cu 1/3 To 1/3	Cu 1/3 To 1/3	Cu* 1/3 To 1/3	Br 1/3 Le 1/3 To* 1/3	Br 1/3 Le* 1/3 To* 1/3	Br 1/3 To* 1/3	Br 1/3	Br* 1/3 Car 1/3 SC 1/3	Br* 1/3 Car 1/3 SC 1/3
Plot 1	98.8 m²		DB Full	DB Full	DB Full	DB Full	DB* Full	Oa Full	Oa Full	Oa Full	Oa Full	Oa Full	Oa Full
Plot 2	123.1 m²	Bu Full	Bu Full	Bu Full	Bu Full	Bu* Full	Bu* Full	BB Full	BB Full	BB Full	BB Full	BB* Full	BB* Full
Plot 3	108.7 m²			Ma Full	Ma Full	Ma Full	Ma Full	Ma Full	Ma* Full				
Plot 4	96.7 m²											Po Full	Po Full

Crop codes

Read left to right from Oct. A star marks the months a crop is being picked; nursery dates and exact spacing are in the full plan.

Leafy crops and herbs

Br = Broccoli

Ca = Cabbage

Co = Coriander

Ka = Kale

Le = Lettuce

SC = Swiss chard (spinach)

TS = True spinach (English spinach)

Roots and onions

Be = Beetroot

Car = Carrots

On = Onions

Po = Potato

SP = Sweet potato

Legumes

BB = Broad beans (fava beans)

DB = Dry beans (sugar beans)

GB = Green beans

Pe = Peas

Fruiting crops

Bu = Butternut

Cu = Cucumber

Pep = Peppers

To = Tomatoes

Staples

Ma = Maize (mielies)

Oa = Oats (winter cover crop)

Original page 5: readable when enlarged, but its crop-cell text is only 5.8 pt in the renderer. The A4 sheet compresses crop codes, fractions and picking marks into a dense matrix. [lib/crop-export-pdf.ts:941](#). The full-plan and field-sheet rows use 8 pt; caveats are smaller and muted.

Three concrete improvements

Continuation layout: page 11 carries only two plot rows; page 23 carries one March harvest row. Page 23 repeats column headings but loses the Harvest and record section heading, so that isolated page is weak context for a field worker.

Reading order: ground preparation follows sowing in the monthly table. Put prep and bed-clearance checks before establishment, so the printed sequence matches the work.

Records: the final monthly record lacks a clear month/year and responsible-person field. Kitchen, stored, sold/shared and loss columns need units; benchmark is a crop-cycle figure and needs a cohort reference before comparison with a monthly actual.

The 16 regional PDFs generated successfully (397 pages total). Automated bounds checks found no out-of-page words on those pages; all 16 food-availability pages were also inspected as rendered images. I did not visually inspect every regional page. Larger paper or a compact field-only pack would improve usability, but neither resolves the missing allocation and timing semantics.

Fix order and acceptance checks

No.	Change to make	What must be visible in the next export
1	Retain all designed food sources, including unknown timing	Banana, hives and coop inventory present; each missing month bar has a reason.
2	Expose animal purpose choices in Simple mode; export undated honey	Honey survives PDF conversion; a coop choice can be set without changing interface mode.
3	Resolve local avocado timing without national union inflation	A cultivar/region-specific season or explicit local-season unknown; bearing/proposed status clear.
4	Use one dated first-year horizon across the pack	October fresh harvest agrees with October tasks; cover, chart, calendar and storage all agree.
5	Keep allocation in every field job	Each sow/transplant row shows its share/area and field strip reference.
6	Confirm whether expired starters happened	Skipped September sowings do not produce November harvests or suppress a replacement purchase.
7	Complete rate-based procurement and storage instructions	Oats shopping row; seed-before-nursery reminders; every stored crop's existing sourced conditions printed.
8	Improve pagination, type and record fields	No orphan action page; section context repeats; month, cohort, units and staff fields present.

Expired starters need an observed state

A focused control created a September green-bean starter and advanced to October without recording any completed sowing. `settleOnceRows` changed it to `existing:true` and the food builder still predicted November picking. The same promotion affects the September starter cohorts mentioned in this export. Aging the row out later does not make that intervening harvest true.

Resolve planted, skipped and overdue separately. Calendar passage can establish that a planned date passed; it cannot establish that a farmer planted. The export should clearly distinguish observed existing crops from scheduled one-offs and repeating future crops.

Tests needed for these fixes

Use exact element IDs and both item statuses; empty and valid animal choices; Simple-mode reachability; banana with an empty sourced season; undated honey through actual PDF export; region/cultivar avocado cases; first-year versus established horizons; missed starters at a month boundary; partial-bed allocations on monthly sheets; rate-based covers in the buying list; and storage-condition text in the exported PDF. Render the corrected Ubhejane pack and compare it with this captured baseline. Tests alone cannot confirm its visual result.

No change has been made to species names, lesson content, saved geometry, `PLAN_VERSION` or app source. This report is a fix brief with reproduced evidence; it does not claim these defects are already repaired.

Evidence, sources and limits

Checked code and verification

Current origin/main snapshot f76814b2aa25 was audited in an isolated managed worktree; the original workspace branch was an older August checkout. Required checks ran in order on Node 24.19.0: `npx tsc --noEmit`, `npm test`, `git diff --check`. Typecheck and whitespace checks passed. The test runner reports 4,380 tests: 4,379 pass, zero fail, one TODO.

The TODO at `tests/shape-sync-loss.test.ts:59` demonstrates a known race where one browser's whole-collection shape push can discard a shape drawn before its listener update. This is outside the crop-season calculation, but it limits confidence that cross-browser saved design geometry is always preserved. I did not change or remove that test.

Evidence delivered alongside this report

`probe.json` reconstructs all 71 printed rows and reproduces the benchmark and task counts; `extra.json` compares dated versus established availability and occupancy; `regional-audit.json` records all 270 scenarios and controls; `animals-audit.json` records the hive/coop cases. The corresponding `.mjs` probes, test log, 16-PDF generation log, PDF geometry checks, and all 30 original page images are retained. The two focused animal PDFs reproduce the export omission. Regional PDFs are bundled separately.

Primary references used to check the concerns

[1] KZN DARD: Fruit and Nut Production in KZN. Avocado cultivar windows, bearing and site requirements; banana cycle and climate/water constraints. This is regional reference guidance, not a verified Ubhejane forecast.

[2] Department of Agriculture / ARC: Cultivation of Avocados; SAAGA industry overview. The code dossier uses their warm/cool Hass windows and national industry span.

[3] SAPA: Commercial Layers training notes. The all-year layer window is conditional on managed lighting, feeding and flock management; it does not establish year-round produce from every smallholder coop.

Honey's local-flow note and records are from the app's cited SANBI Strelitzia 37 dossier. Its SANBI repository link could not be retrieved in this audit, so the underlying honey-flow statements were not independently reverified. No local honey harvest months were inferred.

What was not verified on the actual farm

Private saved design records and enterprise choices; the exact banana element/species selection; orchard cultivar, age and current bearing; whether September starters were planted; irrigation volume/reliability; observed soil, drainage and frost; planting strips on the ground; actual labour and creche meal demand. The printed bed shares and rounded plot areas were reconstructed; some internal flags were inferred. Those assumptions are explicit in `probe.json`.

The live browser exercise used the prepared tour/sample farm. Auto-suggest produced 17 additions and the calendar displayed them; the avocado detail showed the long national season. Reload restored the tour fixture, so this is not proof of persistence in Rory's private saved plan. No paid render, private-data upload, production edit or deployment was performed.